

Users > moshika > Desktop > lab > 🏠 planets.pl

```
1  % --- Facts: planet(Name, DistanceFromSun, Diameter, Moons, Type) ---
2  % Distance in million km, Diameter in km
3
4  planet(mercury, 58, 4879, 0, rocky).
5  planet(venus, 108, 12104, 0, rocky).
6  planet(earth, 150, 12742, 1, rocky).
7  planet(mars, 228, 6779, 2, rocky).
8  planet(jupiter, 778, 139820, 79, gas_giant).
9  planet(uranus, 2871, 50724, 27, ice_giant).
10 planet(saturn, 1430, 116460, 83, gas_giant).
11 planet(neptune, 4495, 49244, 14, ice_giant).
12
13 % --- Rule: get info about a planet ---
14 planet_info(Name) :-
15     planet(Name, Distance, Diameter, Moons, Type),
16     write('Planet: '), write(Name), nl,
17     write('Distance from Sun (million km): '), write(Distance), nl,
18     write('Diameter (km): '), write(Diameter), nl,
19     write('Number of Moons: '), write(Moons), nl,
20     write('Planet Type: '), write(Type), nl.
21
22 % --- Find rocky planets ---
23 rocky_planet(Name) :-
24     planet(Name, _, _, _, rocky).
25
26 % --- Find gas giants ---
27 gas_giant_planet(Name) :-
28     planet(Name, _, _, _, gas_giant).
29
```

```
[moshika@Moshikas-MacBook-Air ~ % cd /Users/moshika/Desktop/lab  
[moshika@Moshikas-MacBook-Air lab % swipl planets.pl  
Welcome to SWI-Prolog (threaded, 64 bits, version 9.2.9)  
SWI-Prolog comes with ABSOLUTELY NO WARRANTY. This is free software.  
Please run ?- license. for legal details.
```

```
For online help and background, visit https://www.swi-prolog.org  
For built-in help, use ?- help(Topic). or ?- apropos(Word).
```

```
[?- planet_info(earth).  
Planet: earth  
Distance from Sun (million km): 150  
Diameter (km): 12742  
Number of Moons: 1  
Planet Type: rocky  
true.
```